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Association between AI-Based Electrocardiographic Age from Wearable Devices and Atrial Fibrillation: The PROPHECG-Age Single Study

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Abstract

Background and Aims: Artificial-intelligence (AI)-derived electrocardiographic (ECG) age is a promising marker of atrial fibrillation (AF) risk, yet it has been evaluated only in hospital-based 12-lead recordings. We aimed to develop PROPHECG-Age Single—an AI model that estimates ECG-age from wearable single-lead ECGs—and to examine whether the resulting ECG-age is associated with AF risk in a real-world self-monitoring setting.

Methods: One million 12-lead ECGs (academic tertiary hospital, Jan 2006–Sep 2021) were converted into synthetic single-lead data via a pre-trained Cycle-Consistent Generative Adversarial Network and used to train a ResNet-1D age-prediction network. The age-prediction model was validated in the S-Patch registry (1,980 participants; Sep 2021–Aug 2024; NCT05119725) and externally in the Memo Patch registry (582 participants; Sep 2022–Nov 2023; NCT05355948). Multivariable logistic (AF presence) and fractional-logit (AF burden) models, adjusted for sex, age, and comorbidities, generated cohort-specific effect estimates that were pooled with fixed-effect meta-analysis.

Results: PROPHECG-Age Single achieved mean absolute errors of 10.01 years (S-Patch) and 11.88 years (Memo Patch). Participants with AF demonstrated significantly larger AI-ECG age gaps than those without AF (–1.2 vs –4.1 years; $p < 0.001$), a difference that persisted after adjustment (odds ratio 1.02 per year; 95% CI 1.01–1.04). Each additional year of AI-ECG age gap showed a 0.74 percentage-point increase in AF burden ($p = 0.030$) after adjustment. Meta-analysis confirmed significant associations with both AF presence (pooled adjusted OR = 1.03 per year; 95% CI 1.01–1.04) and AF burden (pooled marginal effect = 0.008 per year; 95% CI 0.002–0.014).

Conclusions: PROPHECG-Age Single provides ECG-age estimates from wearable devices and robustly associates with AF presence and burden. Wearable-based AI-ECG age is a potential digital biomarker for proactive cardiovascular monitoring in a patient-centred context.

Introduction

Physiological aging is a fundamental contributor to the onset and progression of cardiovascular diseases, notably atrial fibrillation (AF), [1, 2] whose global prevalence continues to rise significantly. [3, 4] Electrocardiography (ECG), a non-invasive and broadly accessible diagnostic tool, captures the electrical activity of the heart and has recently emerged as an intriguing digital biomarker for evaluating cardiovascular aging. Advances in artificial intelligence (AI) now allow precise estimation of an individual's "electrocardiographic age" (AI-ECG age) during sinus rhythm from standard 12-lead ECG recordings. This AI-derived biomarker is especially insightful because discrepancies between AI-predicted ECG age and chronological age—termed the "AI-ECG age gap"—have been strongly correlated with worsening cardiovascular health, heightened mortality, and increased AF risk, including recurrence post-intervention. [5–9]

However, existing AI-ECG age models have primarily relied on episodic, hospital-based 12-lead ECG recordings. Such snapshots inherently miss the continuous electrophysiological variations occurring in everyday life, potentially overlooking subtle yet clinically relevant changes in cardiac aging. Consequently, the episodic nature of these assessments limits the temporal resolution and real-world applicability of AI-derived age metrics. Recent developments in wearable single-lead ECG technology offer promising opportunities for continuous, longitudinal cardiac monitoring outside clinical settings. Yet, a major barrier has been the limited availability of large-scale, high-quality single-lead ECG datasets necessary to train robust AI models suited for continuous wearables. [10]

To overcome this critical limitation, we introduce the PROPHECG-Age Single (PRediction Of PHenotypes using ElectroCardioGraphy-Age Single) model—a novel deep-learning framework utilising large-scale synthetic data specifically designed to estimate AI-ECG age from wearable single-lead ECG recordings within everyday self-monitoring contexts. Through this study, we assess the potential of continuous single-lead AI-ECG age estimation as a transformative digital biomarker, shifting wearable self-monitoring from episodic event detection toward precision-driven, proactive AF risk management.

Methods

Data sources and study populations

This analysis integrated three complementary datasets: (1) Severance Hospital ECG Archive: a retrospective dataset consisting of 1,008,566 high-quality, 12-lead ECGs (10 s duration, 500 Hz sampling rate) obtained from 837,666 adult patients between January 2006 and September 2021. This dataset was exclusively utilized for model training. [7] (2) S-Patch Registry (ClinicalTrials.gov: NCT05119725): a prospective cohort of 1,980 adults recruited across 15 Korean centres from September 2021 to August 2024. Single-lead ECG recordings (S-Patch EX, Wellysis) were continuously collected for up to 72 hours, of which the first consecutive 48-hour segment (starting at 00:00 h) was analyzed. (3) Memo-Patch Registry (ClinicalTrials.gov: NCT05355948): an external validation cohort of 582 adults enrolled at 13 Korean centres between September 2022 and November 2023, from whom single-lead ECG data (MEMO Patch, HUINNO) were recorded for up to 14 days.

All ECG signals underwent a standardized pre-processing pipeline. [11] Following initial band-pass filtering and artifact removal, the highest-quality 10-second ECG segment within every five-minute epoch was selected for analysis. This study was approved by the Institutional Review Board (IRB) of the Yonsei University Health System (IRB No. 4-2024-1455) and adhered to the principles of the Declaration of Helsinki. The IRB waived the requirement for informed consent for this study due to the use of de-identified data and the minimal-risk nature of the research. The detailed descriptions of datasets and preprocessing methodologies are available in Supplementary Method S1.

Model development and training

We constructed the AI model through a two-step deep-learning workflow. Initially, a Cycle-Consistent Generative Adversarial Network (CycleGAN) [12] was developed to translate standard 12-lead ECG recordings (source domain) into synthetic single-lead waveforms that closely matched the signal characteristics of the wearable S-Patch device (target domain). To train this CycleGAN, we utilized a subset of 50,000 randomly selected 12-lead ECGs from the Severance dataset and 100,000 single-lead ECG segments from the S-Patch registry. Once trained, the CycleGAN was applied to the entire Severance dataset (n = 1,008,566 ECGs), generating synthetic single-lead ECG recordings.

Subsequently, these synthetic single-lead ECG recordings were utilized as inputs to train a one-dimensional ResNet model ("PROPHECG-Age Single"), which was adapted from our previously validated PROPHECG-Age architecture. [7] PROPHECG-Age Single was specifically designed to predict electrophysiological ("AI-ECG") age from 10-second single-lead ECG segments sampled at 200 Hz. Comprehensive details regarding the CycleGAN architecture, training hyperparameters, data preprocessing steps, and model optimization strategies are provided in Supplementary Methods S2, with an overview of data curation and model workflow illustrated in Fig. 1.

Validation of AI-ECG Age Estimation in Wearable Single-Lead ECG Cohorts

The PROPHECG-Age Single model was validated in two independent wearable single-lead ECG cohorts (S-Patch and Memo-Patch registries), each employing distinct ECG devices and comprising participants with and without AF.

Internal validation (S-Patch registry): Given that the CycleGAN was trained using single-lead waveforms specifically from the S-Patch device, validation within the S-Patch registry constituted an internal validation. Of 1,980 enrolled participants, 1,502 satisfied analytic inclusion criteria: age 20–90 years, continuous recording ≥ 48 hours, ≥ 1 sinus-rhythm epoch, and available AF status. From each 48-hour recording, the highest-quality 10-second sinus-rhythm ECG segment was sampled every five minutes (maximum 576 segments per participant). Segments failing predefined quality thresholds were discarded, and AI-ECG age was computed as the mean predicted age from the remaining segments. Model accuracy was quantified by mean absolute error (MAE) and Pearson correlation coefficient (r) compared to participants' chronological ages.

External validation (Memo-Patch registry): External validation utilized the Memo-Patch registry. Of 582 initially enrolled participants, 529 met similar analytic inclusion criteria (age 20–90 years, ≥ 1 sinus-rhythm epoch, and available AF status). Participants wore the Memo-Patch device either continuously for 7–14 days (n = 280) or for 24 hours (n = 302). All available ECG data were processed with the same analytical pipeline as the internal cohort: the highest-quality 10-second sinus-rhythm segments extracted every five minutes, removal of segments not meeting quality standards, and calculation of AI-ECG age as the mean of the remaining segments. Model performance in this external cohort was likewise assessed by MAE and Pearson's r with chronological age.

Association of AI-ECG Age Gap with AF Status and AF Burden.

The clinical relevance of the AI-ECG age gap was further investigated in the S-Patch cohort, internally and memo-patch externally. First, we assessed the associations between the AI-ECG age gap during sinus rhythm and AF status (n = 1,502, S-patch). The mean AI-ECG age gap was calculated from sinus rhythms and compared between participants with and without AF using multivariable logistic regression, adjusted for sex and all components of the CHARGE-AF risk score [13], including age, height, weight, blood pressure, smoking status, hypertension, diabetes mellitus, heart failure, and myocardial infarction. An identical modelling approach was applied to the external Memo-Patch cohort (n = 529), and cohort-specific odds ratios per 1-year increase in AI-ECG age gap were pooled via meta-analysis using fixed- or random-effects methods, depending on inter-study heterogeneity.

We further investigated whether the magnitude of the AI-ECG age gap correlated with AF burden among those in whom AF was detected. In the S-Patch cohort, 233 participants had at least one AF episode during the first 48 h, allowing us to define AF burden as the percentage of total recording time spent in AF (range 0–100%). Unadjusted relationships were first explored with Pearson's correlation coefficient. To account for the bounded nature of the AF burden outcome, we then employed fractional logit regression (binomial family, logit link), with AF burden as the dependent variable and AI-ECG age gap as the independent variable, adjusting again for sex and all CHARGE-AF covariates. Model results are presented as average marginal effects, representing the change in percentage points of AF burden per 1-year increment in AI-ECG age gap. In the Memo-Patch cohort, 24 participants with recorded AF episodes underwent the same fractional logit analysis. Cohort-specific marginal effects were subsequently combined through fixed- or random-effects meta-analysis in accordance with observed heterogeneity. Detailed modelling procedures are provided in Supplementary Methods S3.

AI-ECG Age Gap and the Temporal Consistency

To assess the temporal consistency of the AI-ECG age gap as a stable, personalised biomarker, we analysed 214 non-AF participants from the Memo-Patch registry who underwent continuous single-lead ECG monitoring for at least seven days. Each recording was divided into six consecutive 48-hour epochs, and for each epoch we computed the mean AI-ECG age gap. Continuous reproducibility was then quantified in two ways: first, by calculating Pearson's correlation coefficients between the mean age gaps of adjacent epochs to gauge short-term stability; and second, by estimating a two-way mixed-effects intraclass correlation coefficient (ICC) for absolute agreement across all six epochs to capture overall repeatability. To evaluate categorical reliability, we dichotomized the age gap at − 7.5 years—the mean gap observed during the first 48-hour epoch in this non-AF sample—and determined pairwise percentage agreement and Cohen's κ for every possible epoch pair. Together, these analyses demonstrate whether an individual's AI-ECG age gap remains consistent over one week of monitoring.

Statistical Analysis

All quantitative variables were summarised as mean ± SD or median (interquartile range) according to distributional normality, and categorical variables as counts (percentages). Between-group comparisons of continuous outcomes employed Welch's t-test, with multivariable linear regression used for adjusted analyses; categorical differences were tested by χ^2 or Fisher's exact test, followed by multivariable logistic regression when adjustment was required. Within-subject reproducibility of the AI-ECG age gap was quantified with ICC(A,1). Statistical analyses were conducted in Python (NumPy, pandas, SciPy) and R v4.4.3, with meta-analyses performed using the meta package v6.5-0. A two-sided $p < 0.05$ was deemed statistically significant.

Results

Clinical Characteristics of Registry Participants.

Baseline demographic, clinical, and ECG-derived metrics for participants from the S-Patch and Memo-Patch registries are summarized in Table 1. The S-Patch cohort had a notably higher proportion of participants with AF compared to Memo-Patch (81% [n = 1,217] vs. 5% [n = 25]; P < 0.001). Although S-Patch participants were younger (62.2 ± 11.0 vs. 67.4 ± 9.6 years; P < 0.001), their AI-ECG–predicted age at sinus rhythm was paradoxically higher (60.4 ± 9.3 vs. 58.2 ± 8.8 years; P < 0.001), resulting in a higher AI-ECG age gap (− 1.8 ± 12.4 vs. −9.2 ± 11.0 years; P < 0.001). Clinically significant differences included a higher prevalence of congestive heart failure among S-Patch participants (13% vs. 5%; P < 0.001), whereas Memo-Patch users exhibited higher rates of hypertension (66% vs. 55%; P < 0.001) and diabetes mellitus (27% vs. 18%; P < 0.001), resulting in a correspondingly higher CHA_2DS_2-VASc risk score (2.9 ± 1.1 vs. 2.0 ± 1.4; P < 0.001). Additionally, substantial differences were observed in demographic characteristics: the S-Patch registry was predominantly male (66% vs. 29%; P < 0.001) and had a higher proportion of severe alcohol consumption (22% vs. 6%; P < 0.001).

Table 1
Baseline demographic, clinical and ECG-derived metrics of participants in the S-Patch and Memo Patch registries.

	S-patch		**Memo-patch**		
	N	Mean (std)/ N (%)	N	Mean (std)/ N (%)	P-value
Total Sample Count	1502		529		
Atrial fibrillation status	1502		529		**< 0.001**
None		285 (19%)		505 (95%)	
Atrial fibrillation		1217 (81%)		24 (5%)	
AI-ECG Age	1502		529		
AI-ECG Age at sinus rhythm		60.4 (9.3)		58.2 (8.8)	**< 0.001**
AI-ECG Age Gap at sinus		-1.8 (12.4)		-9.2 (11.0)	**< 0.001**
Demographics					
Age	1502	62.2 (11.0)	529	67.4 (9.6)	**< 0.001**
Sex	1502		529		**< 0.001**
Male		993 (66%)		152 (29%)	
Female		509 (34%)		377 (71%)	
BMI	1500	24.5 (4.0)	524	24.2 (3.4)	0.052
Blood pressure	1500		528		
Systolic Pressure (mmHg)		126.8 (15.2)		130.5 (16.5)	**< 0.001**
Diastolic Pressure (mmHg)		74.4 (11.2)		73.5 (11.3)	0.127
Comorbidities	1500		528		
Congestive HF		194 (13%)		25 (5%)	**< 0.001**
Hypertension		828 (55%)		347 (66%)	**< 0.001**
Diabetes mellitus		272 (18%)		144 (27%)	**< 0.001**
Previous Stroke/TIA		146 (10%)		47 (9%)	0.636

Data are presented as mean ± SD for continuous variables and n (%) for categorical variables. P-values refer to comparisons between S-patch and memo-patch registries (Welch t-test or χ^2 test, as appropriate). AI-ECG age refers to the model-predicted “electrophysiological” age; AI-ECG age gap is the difference between AI-ECG age and chronological age. AF burden denotes the percentage of monitoring time spent in atrial fibrillation.

	S-patch		Memo-patch		
CHA2DS2-VASc	1500	2.0 (1.4)	528	2.9 (1.1)	**< 0.001**
HASBLED	1499	1.2 (0.9)	NA	NA	
Smoking status	1500		527		**0.011**
No		1186 (79%)		446 (84%)	
mild		175 12%)		51 (10%)	
Severe		139 (9%)		30 (6%)	
Drinking status	1500		527		**< 0.001**
No		1094 (73%)		436 (82%)	
Mild		71 (5%)		57 (11%)	
Severe		335 (22%)		34 (6%)	

Data are presented as mean ± SD for continuous variables and n (%) for categorical variables. P-values refer to comparisons between S-patch and memo-patch registries (Welch t-test or χ^2 test, as appropriate). AI-ECG age refers to the model-predicted "electrophysiological" age; AI-ECG age gap is the difference between AI-ECG age and chronological age. AF burden denotes the percentage of monitoring time spent in atrial fibrillation.

Abbreviations: AF, atrial fibrillation; BMI, body mass index; HF, heart failure; CHA_2DS_2-VASc, Congestive heart failure, Hypertension, Age ≥75 (doubled), Diabetes mellitus, Stroke/transient ischemic attack (doubled), Vascular disease, Age 65–74, Sex category (female); HAS-BLED, Hypertension, Abnormal renal/liver function, Stroke, Bleeding history, Labile INR, Elderly, Drugs/alcohol concomitantly.

Model Performance

The CycleGAN architecture successfully generated high-fidelity single-lead ECG signals from standard 12-lead recordings, closely matching actual waveforms acquired from wearable S-Patch single-lead devices (Supplementary Figure S1). Training the PROPHECG-Age Single model using these synthetic single-lead ECGs resulted in a mean squared error (MSE) of 203.4 in the training set and 215.9 in the internal validation split, corresponding to a mean absolute error (MAE) of 11.15 years (Supplementary Figure S2).

In real-world wearable ECG data, the PROPHECG-Age Single algorithm maintained robust predictive accuracy. Internal validation within the S-Patch cohort (n = 1,502) demonstrated an MAE of 10.01 years and a significant correlation with chronological age (Pearson's r = 0.26; P < 0.001; Fig. 2A). External validation in the independent Memo-Patch cohort (n = 529) yielded a slightly higher but comparable MAE (11.88 years) with significant correlation (Pearson's r = 0.30; P < 0.001; Fig. 2B).

By comparison, an alternative workflow—in which single-lead wearable ECG data were first reconstructed into 12-lead ECGs using CycleGAN and then processed with the original 12-lead PROPHECG-Age model—achieved a similar MAE (8.86 years) but exhibited weaker correlation (Pearson's r = 0.13; P < 0.001) and notable regression toward the mean (Supplementary Figure S3).

Association Between AI-ECG Age Gap and Prevalent Atrial Fibrillation

A. Internal validation (S-Patch cohort, n = 1,502).

Participants with AF demonstrated a significantly greater AI-ECG age gap compared to those without AF (−1.2 ± 12.3 vs. − 4.1 ± 12.8 years; P < 0.001; Fig. 3A). In multivariable logistic regression adjusted for sex and all CHARGE-AF covariates, each additional 1-year increment in the AI-ECG age gap independently corresponded to 2% higher odds of prevalent AF (adjusted OR: 1.02; 95% CI: 1.01–1.04; Fig. 3B). Analysis by AF subtype revealed a stepwise increase in AI-ECG age gap from no AF (−4.2 ± 12.8 years) through paroxysmal AF (−1.4 ± 12.3 years) to persistent AF (−0.1 ± 12.3 years; P < 0.001 for trend; Supplementary Table S1). Additionally, dichotomization of the AI-ECG age gap at the cohort mean (−1.8 years) identified an elevated gap as uniquely associated with AF (adjusted OR: 1.76; 95% CI: 1.35–2.30; P < 0.001), without significant associations for other cardiovascular comorbidities (Supplementary Figure S4).

B. External validation (Memo-Patch cohort, n = 529) and Meta-analysis.

The baseline characteristics of the external validation cohort, stratified by the presence of AF, are detailed in Supplementary Table S2. In this external sample, each additional year of AI-ECG age gap conferred a non-significant 3% increase in the odds of prevalent AF (adjusted OR 1.03, 95% CI 0.98–1.09; Supplementary Figure S5A). Pooling both cohorts yielded a common-effect OR of 1.03 per 1-year increment in the AI-ECG age gap (95% CI 1.01–1.04) with no evidence of between-study heterogeneity (I^2 = 0%, p = 0.760; Supplementary Figure S5B). These findings indicate a robust, dose-dependent relation between a higher AI-ECG age gap and the presence of AF.

Association Between AI-ECG Age Gap and Atrial Fibrillation Burden

A. Internal validation (S-Patch cohort, n = 233 with AF episodes).

AI-ECG age gap positively correlated with AF burden (Pearson's r = 0.13; P = 0.048; Fig. 4A). The clinical characteristics of these patients, categorized by AF subtype, were otherwise similar (Supplementary Table S3). After multivariable fractional-logit adjustment for sex and CHARGE-AF covariates, each 1-year increase in AI-ECG age gap was independently associated with a 0.74-percentage-point higher AF burden (average marginal effect: 0.0074; 95% CI: 0.002–0.013; Fig. 4B).

B. External validation (Memo-Patch cohort, n = 19 with AF episodes) and Meta-analysis.

Although the direction of effect was consistent (average marginal effect: 0.023; corresponding to a 2.3-percentage-point AF burden increase per year of age gap), the association did not reach statistical

significance due to the small sample size (95% CI: − 0.008 to 0.054; Supplementary Figure S6A). Combining both cohorts via meta-analysis (heterogeneity $I^2 = 0\%$; $P = 0.34$) produced a significant average marginal effect of 0.008 per year of AI-ECG age gap (95% CI: 0.002–0.014; Supplementary Figure S6B), translating to a 0.8-percentage-point increase in AF burden. Collectively, these findings establish the AI-ECG age gap as a consistent quantitative marker of AF burden severity.

Within-subject reproducibility of AI-ECG Age Gap

To evaluate within-subject temporal consistency, each 7–14 day Memo-Patch recording (n = 214, non-AF participants) was segmented into six consecutive 48-hour epochs. The mean AI-ECG age gap exhibited excellent linear reproducibility between adjacent epochs, with Pearson's correlations ranging from 0.90 to 0.98 (epoch 1 vs. epoch 2: $r = 0.96$; $P < 0.001$; Fig. 5A–B). The overall reliability across all six epochs remained very high (two-way mixed-effects ICC[A,1] = 0.93).

When dichotomized at the cohort mean from epoch 1 (–7.5 years), binary agreement with the baseline epoch was substantial but diminished slightly over the 14-day interval, decreasing from 92% in epoch 2 to 71% by epoch 6. Corresponding Cohen's κ values similarly declined, from 0.84 (epoch 2) to 0.39 (epoch 6), reflecting substantial-to-moderate categorical agreement (Fig. 5C–D).

Discussion

Main findings

In this dual-registry cohort study, we developed and validated PROPHECG-Age Single, a novel AI model that estimates "electrocardiographic age" from wearable single-lead ECG devices using synthetic training data. Our findings demonstrate robust clinical associations: each 1-year increase in the AI-ECG age gap conferred 2% higher odds of prevalent AF (adjusted OR 1.02, 95% CI 1.01–1.04) and a 0.74 percentage-point increase in AF burden after comprehensive adjustment. To advance this field and facilitate broader adoption, we are making both the trained model and its weights publicly available to the research community. These results establish the single-lead AI-ECG age gap as a validated, accessible digital biomarker for continuous AF risk assessment, representing a paradigm shift from traditional episodic, hospital-based evaluations toward personalised, patient-centred cardiovascular monitoring.

CycleGAN-based data augmentation enables robust wearable AI-ECG Age estimation

Single-lead ECG-derived AI development has emerged as a highly promising field for wearable cardiovascular monitoring, yet significant challenges have limited its further development. First, single-lead recordings are inherently noisier than standard 12-lead ECGs and lacks the comprehensive spatial information that multi-lead systems provide across different cardiac regions, fundamentally limiting signal quality and interpretability. Second, unlike decades-accumulated standard 12-lead ECG databases

with millions of recordings, available single-lead datasets remain dramatically smaller with limited patient diversity, and the absence of dominant vendors introduces substantial inter-device heterogeneity. To address these limitations, prior studies have attempted to reconstruct 12-lead ECGs from single-lead inputs before applying existing algorithms. [14, 15] However, these approaches face fundamental limitations by attempting to extrapolate limited information into richer representations, resulting in loss of individual physiological variability and convergence toward population averages, [16, 17] as confirmed by our own experiments showing weaker correlations (r = 0.13 vs. 0.26–0.35) and regression-to-the-mean artifacts.

To our knowledge, this is the first study to take the reverse approach: rather than expanding limited single-lead information, we leveraged the rich information content of established 12-lead ECG archives by transforming them into single-lead formats using forward CycleGAN domain adaptation.[18] This information-preserving strategy circumvents the fundamental extrapolation problem, achieving superior performance for individual-level assessment. While the MAE (10.01–11.88 years) derived from PROPHECG-Age Single is modestly higher than our previous 12-lead studies (4.7–7.9 years) [7], this represents the robust performance for continuous single-lead monitoring, as validated across two prospective cohorts using different vendor devices.

AI-ECG Age gap: validated biomarker for AF substrate and burden assessment

Building on our previous research demonstrating the association between AI-derived ECG age and AF risk in 12-lead settings [7], this study validates that single-lead AI-ECG age gap maintains correlations with both AF presence and burden in continuous monitoring environments. Conventional sinus-rhythm ECG interpretation rarely reveals underlying arrhythmogenic conditions, yet AI algorithms can extract subtle, high-dimensional electrophysiological features—such as changes in P-wave morphology and rhythm regularity—that reflect early or latent atrial pathology even during sinus rhythm and in the absence of overt AF episodes. [19]

Positive relationship between AF burden and AI-ECG age gap implies biological plausibility, aligning with the "AF-begets-AF" paradigm where sustained arrhythmic episodes accelerate atrial remodelling, reflected in heightened electrophysiological aging. [20] This also suggests that continuous single-lead monitoring can capture cumulative atrial remodelling processes that may facilitate early AF substrate detection before overt clinical manifestations.

Patient-centred cardiovascular heart through personalised age assessment

Our model represents a highly patient-centred approach by providing individualized electrophysiological age information—arguably the most intuitive and representative biomarker for cardiac health that patients can readily understand and engage with. While wearable ECG technology significantly enhances ambulatory AF detection, the majority of recorded data, even if in previously diagnosed paroxysmal AF,

reflects sinus rhythm intervals (> 99% of monitored time; median annual AF burden ≈ 0.13%), rendering extensive electrophysiological data clinically underutilized. [21] The PROPHECG-Age Single model capitalizes on these intervals by converting subtle electrophysiological variations into a continuous AI-derived age assessment, offering a personalised measure of AF propensity and disease progression not achievable by conventional episodic rhythm detection methods. By making our model and weights publicly accessible and demonstrating robustness across different vendor devices, we facilitate truly democratised, patient-centred care that transcends institutional and technological barriers, potentially accelerating future developments in personalised cardiovascular monitoring.

Limitations

Our study has notable limitations. First, the cross-sectional design precludes establishing causal relationships between the AI-ECG age gap and AF. Second, despite validation in multicentre registries, both cohorts were predominantly East Asian, necessitating validation in diverse populations to ensure broad generalizability. Third, direct comparison between single-lead and standard 12-lead ECG-based AI-ECG age gaps was not feasible due to the absence of 12-lead ECG data from participants. Fourth, although the model demonstrated comparable accuracy to high-resolution 12-lead approaches, it exhibited modestly higher mean absolute errors attributable to the shorter duration and lower sampling rate of single-lead recordings. Finally, the lack of uncertainty estimates limits precise differentiation of true physiological variability from model prediction error at the individual level.

Conclusions

PROPROPHECG-Age Single successfully transitions electrophysiological age estimation from episodic 12-lead ECGs to continuous, wearable single-lead monitoring. By translating sinus rhythm data into a dynamic AI-derived age gap, this model enhances traditional event-driven detection methods, providing a robust biomarker reflective of AF substrate, burden, and underlying genetic and structural remodelling. Implementing this approach in wearable technology could advance personalized atrial health management, enabling earlier, precision-guided interventions and significantly enhancing AF preventive care at scale.

Declarations

Author Contributions Statement

S.H.P. led the study, taking primary responsibility for manuscript drafting (including tables/figures), AI model development, and data analysis. J.J. contributed to data analysis and provided research assistance. B.Y.J. designed and supervised the cohort studies. S.C.Y. contributed to model development, designed the statistical analysis plan, and provided overall supervision. The study concept was developed jointly by S.C.Y., H.T.Y., and B.Y.J., who also offered critical feedback throughout the project. J.K., D.L., D.K., and J.Ja. (industry collaborators) secured, processed, and provided the single-lead wearable ECG data. All authors reviewed and approved the final manuscript.

Disclosure of Interest

Seung Hyun Park: Nothing to declare beyond institutional funding reported below.

Ju Hyun Jin: Nothing to declare.

Jongwoo Kim: Pending patent applications related to atrial fibrillation prediction using AI (United States Application No. 18/636,402, filed 15 April 2024; Republic of Korea Application No. 10-2023-0069397, filed 30 May 2023). Shareholder of Wellysis Corp.

Dongha Lee: Nothing to declare.

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Boyoung Joung: Nothing to declare beyond institutional funding reported below.

Data Availability

Anonymised data used in this study will be made available to qualified investigators for the purpose of replicating the analyses and findings, subject to appropriate ethical approvals and institutional authorisations. The complete AI algorithm (*PROPHECG-Age Single*), including trained weights, is openly accessible via GitHub at: https://github.com/dr-you-group/PROPHECG-Age-Single. Additional processed data, related materials, and programming code are available from the corresponding authors upon reasonable request.

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Ethical Approval

The study protocol was reviewed and approved by the Institutional Review Board of Severance Hospital, Yonsei University Health System (IRB No. 4-2024-1455).

Pre-registered Clinical Trial Number

The *S-Patch* registry (ClinicalTrials.gov identifier: NCT05119725; 1,980 participants; September 2021–August 2024) and the *Memo-Patch* registry (ClinicalTrials.gov identifier: NCT05355948; 582 participants; September 2022–November 2023) were both pre-registered clinical studies.

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Figures

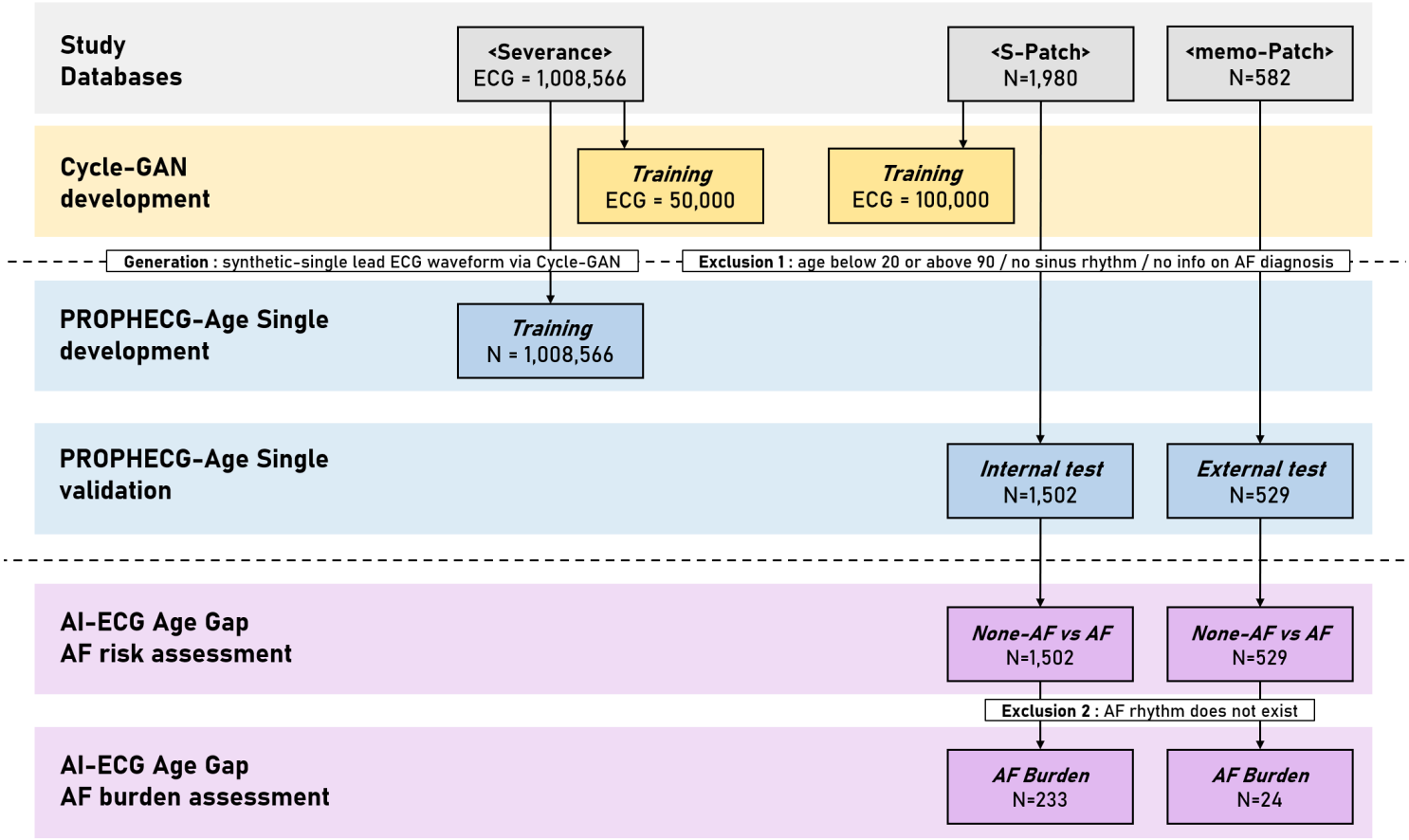

Figure 1

Data Sources, Model Development, and Analytical Cohort Derivation for PROPHECG-Age Single

Schematic representation illustrating the data integration, model training, validation, and downstream clinical analyses. Three distinct datasets were utilized: the Severance Hospital ECG archive (1,008,566 standard 12-lead ECGs), the S-Patch registry (1,980 participants), and the Memo-Patch registry (582 participants). A Cycle-Consistent Generative Adversarial Network (CycleGAN) was first trained on 50,000 standard 12-lead ECGs (Severance) and 100,000 single-lead ECGs (S-Patch), and subsequently generated synthetic single-lead waveforms from the full Severance dataset (n = 1,008,566). These synthetic single-lead waveforms were used to train the PROPHECG-Age Single model, a one-dimensional ResNet architecture designed to estimate electrophysiological (AI-ECG) age from 10-second single-lead ECG segments. Internal validation utilized 1,502 eligible S-Patch participants, whereas external validation employed 529 eligible Memo-Patch participants. For clinical evaluation, analyses of prevalent atrial fibrillation (AF) presence were conducted for all eligible participants (S-Patch, n = 1,502; Memo-Patch, n = 529). Participants demonstrating at least one AF episode underwent additional AF burden analyses (S-Patch, n = 233; Memo-Patch, n = 24).

Abbreviations: AF, atrial fibrillation; ECG, electrocardiogram; CycleGAN, Cycle-Consistent Generative Adversarial Network.

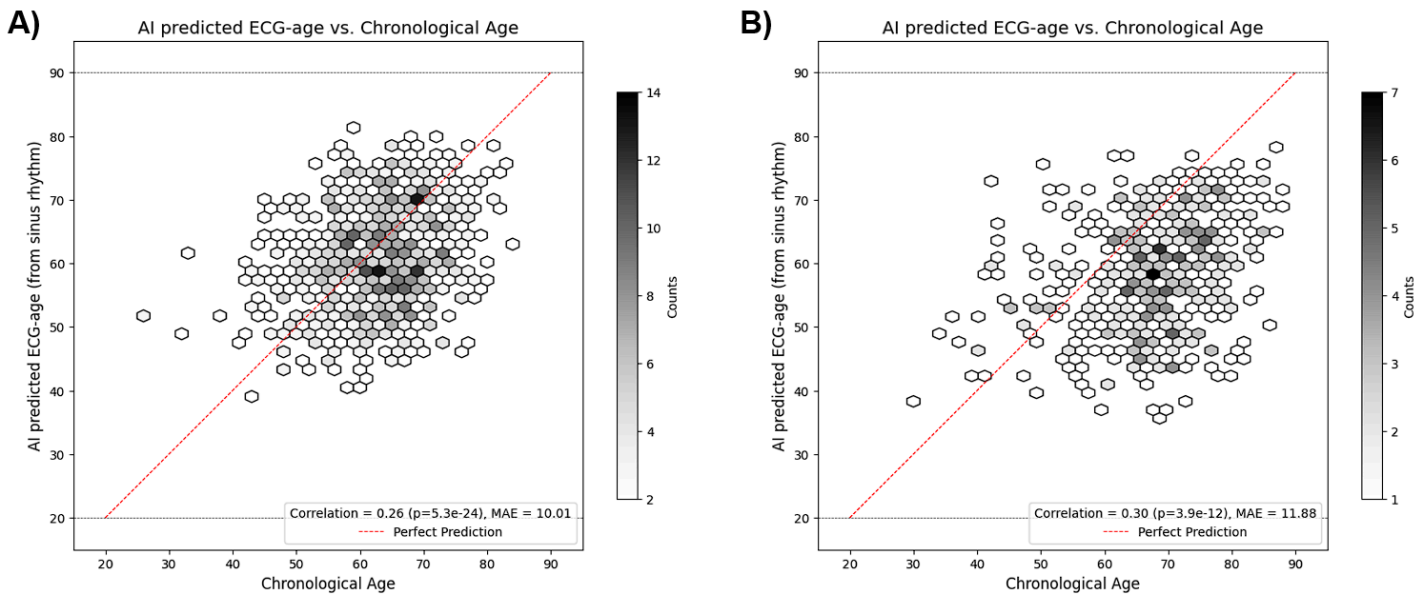

Figure 2

Performance of PROPHECG-Age Single: correlation between AI-predicted ECG age and chronological age

Hexbin scatter plots showing the relationship between AI-predicted ECG age (derived from continuous single-lead, sinus-rhythm segments) and true chronological age in (A) the internal validation cohort (S-Patch registry, n = 1,502; Pearson r = 0.26, p < 0.001; MAE = 10.01 years) and (B) the external validation cohort (Memo Patch registry, n = 529; r = 0.30, p < 0.001; MAE = 11.88 years). The red dashed line indicates perfect prediction (y = x), and darker hexagons denote regions of higher sample density.

Abbreviations: AF, atrial fibrillation; ECG, electrocardiogram

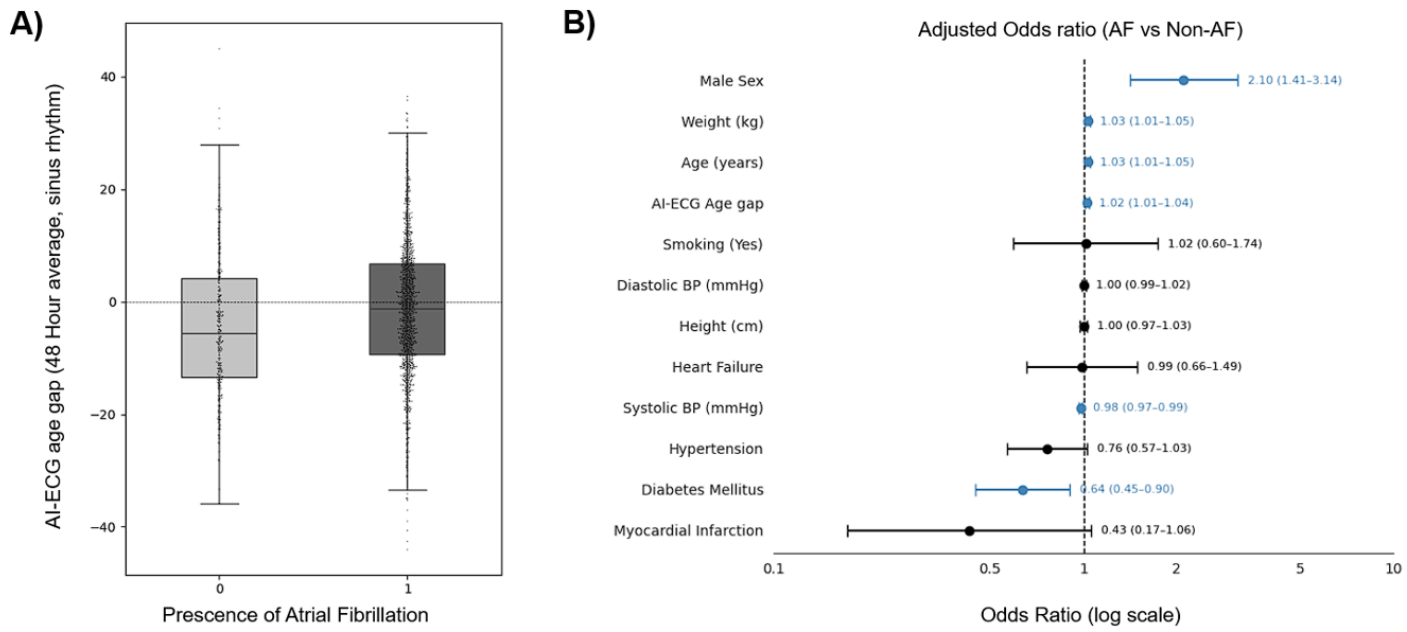

Figure 3

Relationship Between AI-ECG age gap and Atrial Fibrillation Presence

(A) Box-and-whisker plot (with overlaid individual data points) comparing the 48-hour average AI-ECG age gap during sinus rhythm in participants without AF (n = 1,217) versus those with AF (n = 285). The AF group shows a significantly greater positive age gap (Welch's t-test, $p < 0.001$).

(B) Forest plot of adjusted odds ratios (ORs) and 95% confidence intervals (CIs) from a multivariable logistic regression for AF presence. Each 1-year increment in the AI-ECG sinus age gap is associated with an OR of 1.02 (95% CI: 1.01–1.04), indicating a significant independent predictor of AF. Other covariates include male sex, heart failure, weight, chronological age, smoking status, systolic and diastolic blood pressure, height, hypertension, diabetes mellitus, and prior myocardial infarction. The dashed vertical line marks OR = 1; blue markers denote predictors with $p < 0.05$.

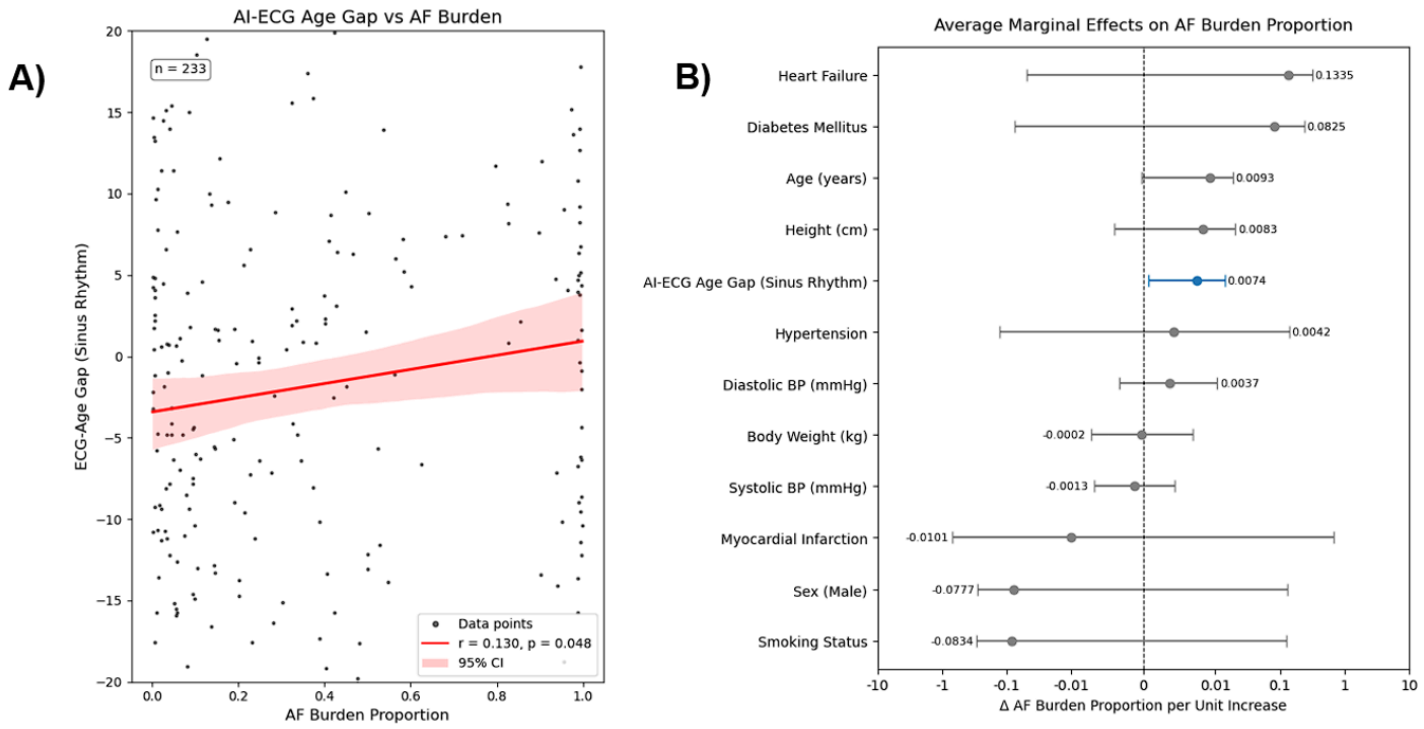

Figure 4

Relationship Between AI-ECG Age Gap and Atrial Fibrillation Burden

This analysis was conducted in all 233 subjects with AF burden > 0% detected by the wearable device (including both device-detected and clinically adjudicated AF episodes).

(A) Box-and-whisker plot (with overlaid individual data points) comparing the 48-hour average AI-ECG age gap during sinus rhythm between participants with lower (≤ 23.9%) versus higher (> 23.9%) AF burden (median split).

(B) Forest plot of adjusted regression coefficients (β) and 95% confidence intervals from a multivariable linear model predicting AF burden (%) as the outcome. The AI-ECG age gap (per 1-year increment) is independently associated with greater AF burden, the average marginal effect of the age gap was 0.0074 (95 % CI 0.002–0.013), indicating that each additional 1-year increase in the gap was associated with a 0.74-percentage-point rise in AF burden Covariates include heart failure, diabetes mellitus, chronological age, height, diastolic and systolic blood pressure, weight, prior myocardial infarction, antihypertensive medication, sex, and smoking status. The dashed vertical line indicates β = 0; blue markers denote significant predictors ($p < 0.05$).

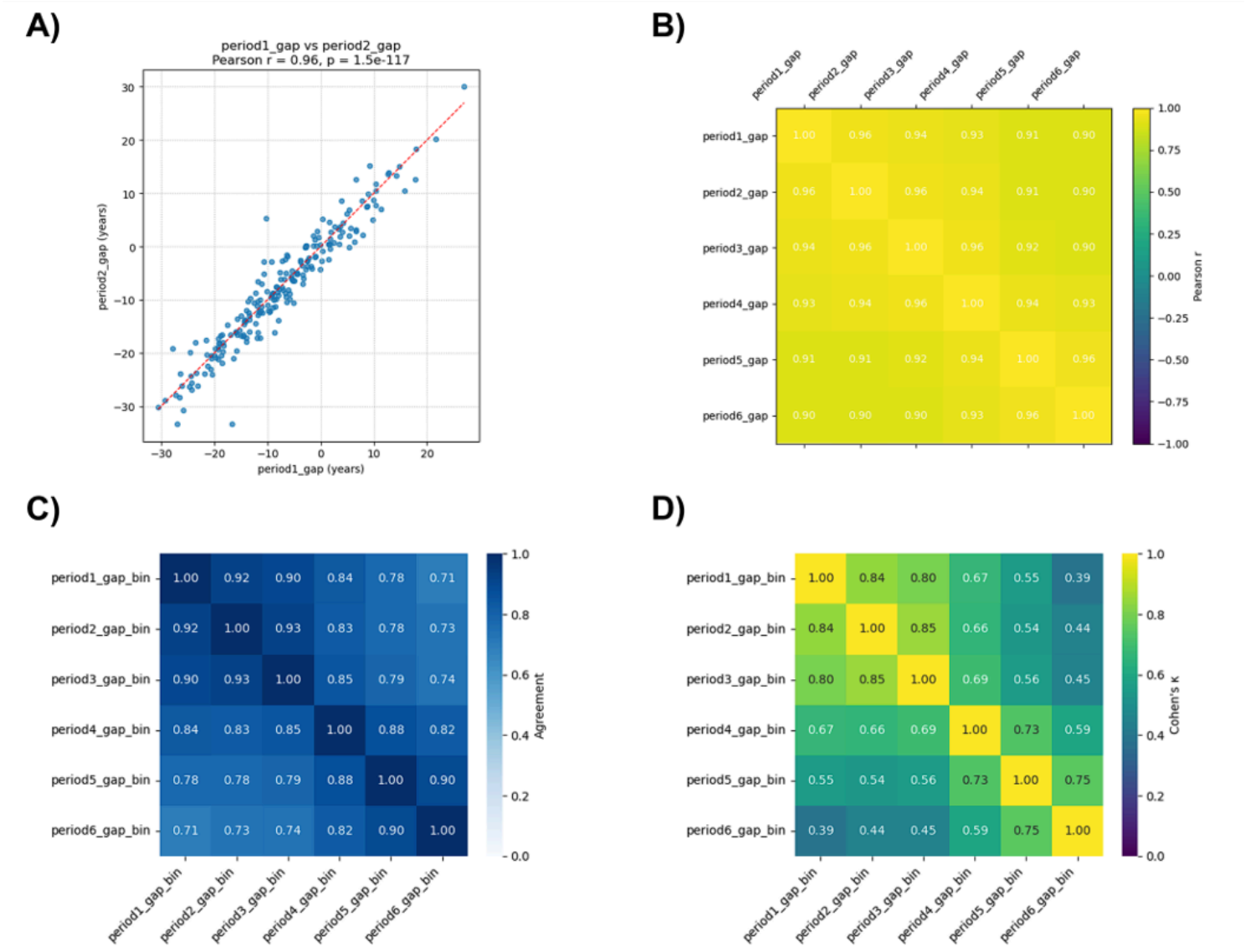

Figure 5

Reproducibility of AI-ECG Age Gap Across Consecutive 48-Hour Monitoring Periods

In the Memo Patch cohort, continuous single-lead ECG recordings were divided into six sequential 48-hour segments (period 1 = first 48 h, period 2 = second 48 h, …, period 6). At each segment, the AI-ECG age gap was computed as (predicted ECG age − chronological age). To assess categorical consistency, each period_gap was binarized using the threshold of −7.5 years (the mean AI-ECG age gap in Memo Patch period 1): “high” if > −7.5 years and “low” if ≤ −7.5 years. (A) Scatter plot of period 1 versus period 2 AI-ECG age gaps (years), demonstrating excellent linear agreement (Pearson r = 0.96, p <0.001, red dashed line = y = x). (B) Pearson correlation matrix for AI-ECG age gaps across all six periods, with values indicating pairwise r. (C) Pairwise simple agreement matrix (proportion concordant) for these binary categories. (D) Pairwise Cohen’s κ matrix quantifying beyond-chance agreement for the same binary classification.

Abbreviations: AI, artificial intelligence; ECG, electrocardiogram; r, Pearson correlation coefficient; κ, Cohen's kappa

Supplementary Files

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- supplemethodsfiguretable0920.docx